

In this paper, we observe that typical manifolds,

We promise that: equivalent relation on X generated by: condition means the smallest equivalent relation on X satisfying the condition.

Rigorously, $\sim := \bigcap R_1 \times R_2$
 $R_1 \times R_2 \in \mathcal{X} \times \mathcal{X}$ is eq-rel
 $R_1 \times R_2$ satisfies the condition

And, $I = [0, 1]$, the subspace of the euclidean space $\mathbb{R}$.

$$\begin{array}{ccc} X & & \\ \pi \downarrow & \searrow f & \\ X/\sim & \xrightarrow{\tilde{f}} & Y \end{array}$$

Cylinder

Def.

Define an equivalence relation $\sim$ on $I \times I$ generated by: $\forall y \in I, (0, y) \sim (1, y)$
The quotient space $I^2/\sim$ is called the Cylinder.

Prop. The cylinder $I^2/\sim$ is Homeomorphic to $S^1 \times I \subseteq \mathbb{C} \times \mathbb{R}$.

Proof. Define a map

$$f: I^2 \rightarrow S^1 \times I: (x, y) \mapsto (e^{2\pi i x}, y)$$

$$\begin{array}{ccc} I^2 & & f \\ & \searrow & \\ \pi \downarrow & & S^1 \times I \\ I^2/\sim & \xrightarrow{\tilde{f}} & \end{array}$$

Then, f is continuous onto closed map, because

1. Continuous

$(x, y) \mapsto e^{2\pi i x}$ and $(x, y) \mapsto y$ are continuous, so f is also continuous.

2. Onto

Given $(e^{i\theta}, b) \in S^1 \times I$ where $\theta \in [0, 2\pi]$, take $x = \frac{\theta}{2\pi}$ and $y = b$, then clear.

3. Closed

Given closed set $C \subseteq I^2$, C is compact since I^2 is compact.

So, the continuous image $f[C]$ is also compact in $S^1 \times I$,

therefore $f[C]$ is closed, by $S^1 \times I$ is Hausdorff.

Moreover, $f(x_1, y_1) = f(x_2, y_2) \Leftrightarrow e^{2\pi i x_1} = e^{2\pi i x_2}$ and $y_1 = y_2$
($\Leftrightarrow e^{2\pi i(x_1 - x_2)} = 1$)

$$\Leftrightarrow \left(x_1 = x_2 \text{ or } \{x_1, x_2\} = \{0, 1\} \right) \text{ and } y_1 = y_2$$

$$\Leftrightarrow (x_1, y_1) \sim (x_2, y_2)$$

Therefore, f induces a Homeomorphism

$$\tilde{f}: I^2/\sim \rightarrow S^1 \times I: [(x, y)] \mapsto f(x, y).$$

It is injective and well-defined by $(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow f(x_1, y_1) = f(x_2, y_2)$

And surjective is given by the f is surjective.

Finally, the continuity is given by: $\tilde{f} \circ \pi = f$ is continuous, and,

given closed set $C \subseteq I^2/\sim$, $\pi^{-1}[C]$ is closed, so the ontoness of π gives

$\tilde{f}[C] = \tilde{f}[\pi[\pi^{-1}[C]]] = f[\pi^{-1}[C]]$ is closed, since f is closed map. $\square$

Torus

Def.

Define an equivalence relation $\sim$ on $I \times I$ generated by:

$$\left\{ \begin{array}{l} \forall x, y \in I, (x, 0) \sim (x, 1) \text{ and } (0, y) \sim (1, y) \\ (0, 0) \sim (1, 0) \sim (0, 1) \sim (1, 1) \end{array} \right.$$

The quotient space $I^2/\sim$ is called a Torus.

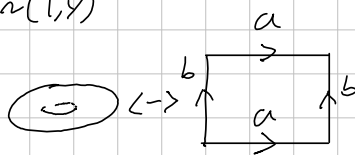
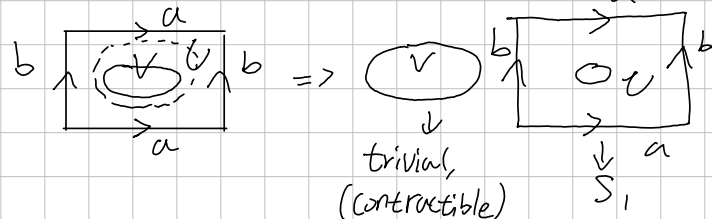
Prop. The Torus $I^2/\sim$ is Homeomorphic to $S^1 \times S^1 \subseteq \mathbb{C}$.

Proof. Define a map

$$f: I^2 \rightarrow S^1 \times S^1: (x, y) \mapsto (e^{2\pi i x}, e^{2\pi i y})$$

Then, f is continuous onto closed map,

Fundamental group of the Torus,



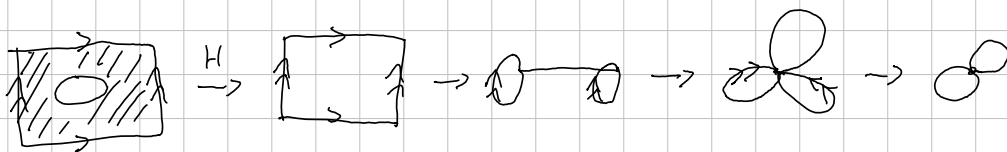
$$\langle a, b \mid aba^{-1}b^{-1} = e \rangle$$

$$\cong \mathbb{Z} \times \mathbb{Z}$$

$I^2/\sim \cong S^1 \times S^1$ gives

$$\pi_1(I^2/\sim) \cong \pi_1(S^1 \times S^1)$$

$$\cong \pi_1(S^1) \times \pi_1(S^1)$$



Therefore, $\pi_1(U) * \pi_1(V) / \pi_1(U \cap V)$

$$\cong \langle a \rangle * \langle b \rangle / \langle aba^{-1}b^{-1} \rangle$$

$$\cong \mathbb{Z} \times \mathbb{Z}.$$

Projective Space.

Def. Define an equivalent relation on $\mathbb{R}^{n+1} \setminus \{0\}$ generated by:

$$x \sim y \Leftrightarrow \exists t \in \mathbb{R} \text{ such that } tx = y.$$

The quotient space $\mathbb{RP}^n \stackrel{\text{def}}{=} (\mathbb{R}^{n+1} \setminus \{0\}) / \sim$ is called a Real Projective space.

(2) $\mathbb{RP}^n$ 은 $S^n = \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\}$ 상의 동치관계

$$x \sim' - x \quad (x \in S^n)$$

에 의한 상공간 $S^n / \sim'$ 과 위상동형임을 다음과 같이 보일 수 있다. 이를 위해 다음과 같은 도표를 생각하자.

$$\begin{array}{ccc} S^n & \xrightarrow{i} & \mathbb{R}^{n+1} - \{0\} \\ \pi \downarrow & & \downarrow \pi \\ S^n / \sim' & \xrightarrow{k} & \mathbb{RP}^n \end{array}$$

여기서 $i: S^n \hookrightarrow \mathbb{R}^{n+1} - \{0\}$, $x \mapsto x$ 는 포함사상이다.

(i) 먼저 연속사상 $\pi \circ i: S^n \rightarrow \mathbb{RP}^n$ 가 전사임을 보이자. 임의의 $[x] \in \mathbb{RP}^n$ 에 대하여 $x \in \mathbb{R}^{n+1} - \{0\}$ 이므로 $\|x\| \neq 0$ 이다. 그러면 $\frac{x}{\|x\|} \in S^n$ 이고 $\frac{x}{\|x\|} \sim x$ 이므로 $\pi \circ i \left(\frac{x}{\|x\|} \right) = \pi \left(\frac{x}{\|x\|} \right) = \pi(x) = [x]$ 이다. 따라서 $\pi \circ i$ 는 전사인 연속사상이다.

(ii) 이제

$$x \sim' y \Leftrightarrow \pi \circ i(x) = \pi \circ i(y)$$

임을 보이자. 만약 $x \sim' y$ 이면 $x \sim y$ 이므로 $\pi \circ i(x) = \pi(x) = \pi(y) = \pi \circ i(y)$ 이다. 역으로 만약 두 점 $x, y \in S^n$ 가 $\pi \circ i(x) = \pi \circ i(y)$ 를 만족하면 $\pi(x) = \pi(y)$ 이므로 적당한 실수 $t \in \mathbb{R} - \{0\}$ 가 존재하여 $y = tx$ 이다. 그런데 $x, y \in S^n$ 이므로 $1 = \|y\| = \|tx\| = |t| \|x\| = |t|$ 이 되어야 하므로 $t = \pm 1$ 이다. 따라서 $y = \pm x$ 이므로 $x \sim' y$ 이다.

(iii) 더 나아가 S^n 이 콤팩트이고 $\mathbb{RP}^n = (\mathbb{R}^{n+1} - \{0\}) / \sim$ 은 하우스도르프이므로, 따름정리 13.1.13에 의해, $\pi \circ i$ 는 위상동형사상 $k: S^n / \sim' \rightarrow \mathbb{RP}^n$ 를 유도한다. 특별히 $n=1$ 일 때는 $\mathbb{RP}^1$ 은 $S^1 \subset \mathbb{R}^2$ 과 위상동형이다. 구체적으로 연속사상

$$f: S^1 \rightarrow S^1, f(\cos \theta, \sin \theta) = (\cos 2\theta, \sin 2\theta)$$

가 위상동형사상 $\tilde{f}: \mathbb{RP}^1 \rightarrow S^1$ 를 유도한다.⁸⁾

(3) 또한 $\mathbb{RP}^n$ 은 $D^n = \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$ 상의 동치관계

$$x \sim'' -x \quad (x \in S^{n-1})$$

Prop. $\mathbb{RP}^1$ is Homeomorphic to the circle S^1

Proof. Define a map

$$f: \mathbb{R}^2 \setminus \{(0,0)\} \rightarrow S^1: (x,y) \mapsto \frac{1}{x^2+y^2} \cdot (x^2-y^2, 2xy)$$

It is derived by $r \cdot e^{i\theta} \mapsto e^{i \cdot 2\theta}$, so f is onto. Moreover, it is continuous map, being rational function, entrywise.

And, observe that

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow \theta_1 - \theta_2 = k\pi \text{ for some } k \in \mathbb{Z} \quad \text{where } e^{i\theta_j} = (x_j, y_j) \quad (j=1,2)$$

$$\Leftrightarrow e^{i \cdot 2(\theta_1 - \theta_2)} = 1$$

$$\Leftrightarrow f(x_1, y_1) = f(x_2, y_2).$$

Therefore, the induced map

$$\tilde{f}: \mathbb{RP}^1 \rightarrow S^1: [(x,y)] \mapsto f(x,y)$$

is well-defined, injective, surjective.

Moreover, for $\tilde{\iota}: S^1 \rightarrow \mathbb{R}^2 \setminus \{(0,0)\}: (x,y) \mapsto (x,y)$,

$$\pi \circ \tilde{\iota}: S^1 \rightarrow \mathbb{RP}^1$$

$$\begin{array}{ccc} \mathbb{R}^2 \setminus \{(0,0)\} & & \\ \pi \downarrow & \searrow f & \\ \mathbb{RP}^1 & \xrightarrow{\tilde{f}} & S^1 \end{array}$$

$$f = \tilde{f} \circ \pi.$$

is continuous onto map, and since S^1 is Compact, so $\mathbb{RP}^1$ is Compact. Finally, $\mathbb{RP}^1$ is compact and S^1 is Hausdorff, thus $\tilde{f}$ is closed map, therefore $\tilde{f}$ is also closed map, so $\mathbb{RP}^1 \cong S^1$ $\square$

Covering space of $\mathbb{RP}^n$.

covering map.

Thm. Let $X \sim X$ ($X \in S^2$) be an equivalence relation.
Then, $\mathbb{RP}^2 \cong S^2/\sim$ and the natural Projection

$$\pi: S^2 \rightarrow S^2/\sim (\cong \mathbb{RP}^2): X \mapsto [X]$$

is a Covering map.

Proof. Clearly, π is Continuous onto map. Therefore, enough to show that:
Given $[b] \in \mathbb{RP}^2$, there exists evenly covered open neighbor containing $[b]$.
To show this, notice that $\pi^{-1}([b]) = \{b, -b\}$. Define

$$V_1 = \{X \in S^2 \mid \langle X, b \rangle > \frac{1}{2}\}, \text{ (that is, } V_1 \text{ is a set of all points degree } < \frac{\pi}{3} \text{)}$$

$$V_2 = \{X \in S^2 \mid \langle X, -b \rangle > \frac{1}{2}\} = -V_1. \text{ Then, } V_1, V_2 \subset S^2 \text{ are disjoint open sets.}$$

Now, let $U = \pi[V_1] (= \pi[V_2])$.

$$\text{Then, } \pi^{-1}[U] = \pi^{-1}[\pi[V_1]] = V_1 \cup (-V_1) = V_1 \cup V_2,$$

therefore U is open, since π is quotient map,

and $\pi|_{V_i} \rightarrow U$ ($i=1,2$) are Homeomorphism, because:
Continuous and onto are clear, and given open set $W \subset V_i$,

$$\pi^{-1}[\pi[W]] = W \cup (-W), \text{ open, thus } \pi[W] \text{ is open.}$$

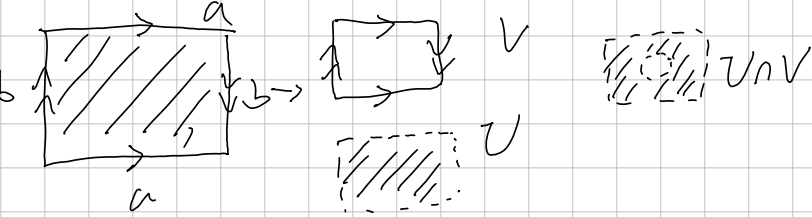
And, given $b_1, b_2 \in V_i$, $\pi(b_1) = \pi(b_2)$ implies that $b_1 = b_2$, because

$$\{b_1\} = [b_1] \cap V_i = [b_2] \cap V_i = \{b_2\}.$$

Consequently, π is a Covering map. $\square$

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Klein bottle.



$$\rightarrow \pi_1(V) = \langle a \rangle * \langle b \rangle$$

$$\pi_1(U) = \{1\}$$

$$\pi_1(U \cap V) = \langle aba^{-1}b \rangle$$

$$\text{so, } \pi_1(K) \cong \langle a \rangle * \langle b \rangle / \langle aba^{-1}b \rangle$$

$$= \langle a, b \mid aba^{-1}b = 1 \rangle$$

Möbius band.

Sphere

Thm. $D^n/S^{n-1} \cong S^n$

Proof. Define a map

$$f: D^n \rightarrow S^n: X = (x_1, \dots, x_n) \mapsto (2x_1 \sqrt{1 - \|x\|^2}, 2x_2 \sqrt{1 - \|x\|^2}, \dots, 2x_n \sqrt{1 - \|x\|^2}, 2(\|x\|^2 - 1))$$

